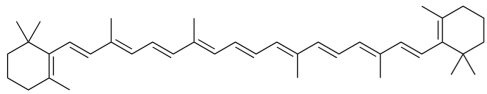
Q1. Assume 22 π electrons of a β-carotene (**1**) move along a straight line whose length can be estimated as 29.4 Å. The ground state energy level has the quantum number $n = 11$. Estimate the absorption wavelength for electronic transition to occur from the ground state to the next excited state.



1 β-Carotene

The length of β-carotene's descendant molecule retinal (vitamin A) is estimated to be 16.8 Å. Show that the first electronic transition is predicted to occur at 1.4×10^{6} m^{-1}. (Retinal has 12 π electrons.)

What happened to the wavelength (λ) of absorption as the length of conjugation decreases from β-carotene to retinal ?

A1

FOR β-Carotene

$$\Delta E = \frac{(2n+1)\,h^2}{8ma^2}$$

$n = 11$

$a = 29.4\ \text{Å} = 29.4 \times 10^{-10}\ m$

$$= \frac{23 \times (6.626 \times 10^{-34}\ J.s)^2}{8 \times 9.109 \times 10^{-31}\ kg \times (29.4 \times 10^{-10}\ m)^2}$$

$$= 1.6 \times 10^{-19}\ J$$

$$\Delta E = h\nu = \frac{hc}{\lambda} = hc\bar{\nu}$$

$$\Rightarrow \bar{\nu} = \frac{\Delta E}{hc} = \frac{1.6 \times 10^{-19}\ J}{6.626 \times 10^{-34}\ J.s \times 2.99 \times 10^{8}\ m/s}$$

$$= \frac{0.8 \times 10^{6}}{m} = \frac{0.8 \times 10^{6}}{100\ cm}$$

$$= 0.8 \times 10^{4}\ cm^{-1}$$

$$\lambda = \frac{1}{\bar{\nu}}$$

$\bar{\nu}_{beta} = 0.8 \times 10^{4}\ cm^{-1}$

$\lambda_{beta} \approx 1241\ nm$

FOR retinal

$$\Delta E = (2n+1)\frac{h^2}{8ma^2}$$

$n = 6$

$a = 16.8\ \text{Å} = 16.8 \times 10^{-10}\ m$

$$= \frac{13 \times (6.626 \times 10^{-34}\ J.S)^2}{8 \times 9.109 \times 10^{-31}\ kg \times (16.8 \times 10^{-10}\ m)^2}$$

$$= 2.8 \times 10^{-19}\ J$$

$$\Delta E = h\nu = \frac{hc}{\lambda} = hc\bar{\nu}$$

$$\Rightarrow \bar{\nu} = \frac{\Delta E}{hc} = \frac{2.8 \times 10^{-19}\ J}{6.626 \times 10^{-34}\ J.S \times 2.99 \times 10^{8}\ m/s}$$

$$= \frac{1.4 \times 10^{6}}{m} = \frac{1.4 \times 10^{6}}{100\ cm}$$

$$= 1.4 \times 10^{4}\ cm^{-1}$$

$\bar{\nu}_{reti} = 1.4 \times 10^{4}\ cm^{-1}$

$\lambda_{reti} \approx 716\ nm$

$\bar{\nu}_{beta} = 0.8 \times 10^{4}\ cm^{-1}$

$\lambda_{beta} \approx 1241\ nm$

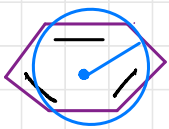
Length of conjugation decreases ↓

Absorption wavelength decreases ↓

Q2. Imagine six π electrons of benzene freely moving in a circular ring of carbon atoms. The carbon-carbon bond length in benzene is 1.4 Å (can be assumed as the radius of the circle). Calculate the absorption wavelength for electronic transition to occur from the ground state (highest occupied) to the next excited state (lowest unoccupied). Your answer should include an energy-level diagram showing the six π electrons accommodated in appropriate energy levels.

A2

3π bonds
6π electrons





FOR benzene

$$\Delta E = \frac{(2m_l+1)\,h^2}{8\pi^2 m r^2}$$

$m_l = 1$

$r = 1.4\ \text{Å} = 1.4 \times 10^{-10}\ m$

$$= \frac{3 \times (6.626 \times 10^{-34}\ J.s)^2}{8 \times \pi^2 \times 9.109 \times 10^{-31}\ kg \times (1.4 \times 10^{-10}\ m)^2}$$

$$= 9.34 \times 10^{-19}\ J$$

$$\Delta E = h\nu = \frac{hc}{\lambda} = hc\bar{\nu}$$

$$\Rightarrow \bar{\nu} = \frac{\Delta E}{hc} = \frac{9.34 \times 10^{-19}\ J}{6.626 \times 10^{-34}\ J.s \times 2.99 \times 10^{8}\ m/s}$$

$$= \frac{4.7 \times 10^{6}}{m} = \frac{4.7 \times 10^{6}}{100\ cm}$$

$$= 4.7 \times 10^{4}\ cm^{-1}$$

$$\lambda = \frac{1}{\bar{\nu}} = 2.12 \times 10^{-7}\ m = 212\ nm$$

Q3. The two-dimensional (planar) rotational motion of a free particle is modeled using the time-independent Schrödinger equation. Two wavefunctions (eigenfunctions) are obtained:

$$\psi_+(\phi) = \frac{1}{\sqrt{2\pi}} e^{im_l\phi} \qquad \text{and} \qquad \psi_-(\phi) = \frac{1}{\sqrt{2\pi}} e^{-im_l\phi}$$

Show that they represent two-fold degenerate states.

A3

For Rotational motion of a free particle

$$\hat{H} = -\frac{\hbar^2}{2I}\frac{d^2}{d\phi^2}$$

$$\hat{H}\psi_+(\phi) = -\frac{\hbar^2}{2I}\frac{d^2\psi_+}{d\phi^2} = -\frac{\hbar^2}{2I}\frac{d^2}{d\phi^2}\left[\frac{1}{\sqrt{2\pi}} e^{im_l\phi}\right]$$

$$= -\frac{\hbar^2}{2I} \times (i)^2 \times (m_l)^2 \frac{1}{\sqrt{2\pi}} e^{im_l\phi}$$

$$= -\frac{\hbar^2}{2I} \times (-1) \times m_l^2 \psi_+ = \frac{\hbar^2 m_l^2}{2I}\psi_+$$

eigenvalue (Total energy)

$$\hat{H}\psi_-(\phi) = -\frac{\hbar^2}{2I}\frac{d^2\psi_-}{d\phi^2} = -\frac{\hbar^2}{2I}\frac{d^2}{d\phi^2}\left[\frac{1}{\sqrt{2\pi}} e^{-im_l\phi}\right]$$

$$= -\frac{\hbar^2}{2I} \times (-i)^2 \times (m_l)^2 \frac{1}{\sqrt{2\pi}} e^{-im_l\phi}$$

$$= -\frac{\hbar^2}{2I} \times (-1) \times m_l^2 \psi_- = \frac{\hbar^2 m_l^2}{2I}\psi_-$$

eigenvalue (Total energy)

Same energies obtained from two wavefunctions → hence → two-fold degeneracies